

Generative Optimization for Incentivized Advertising with Global Level Constraints

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Abstract

Incentivized advertising allocates monetary or virtual rewards to drive user engagement, where a key challenge is optimizing continuous incentive magnitudes under strict global constraints. This problem is complicated by high-frequency interactions, delayed feedback, and non-Markovian user dynamics such as fatigue, which limit the effectiveness of existing uplift modeling and constrained reinforcement learning approaches. To address these challenges, we propose GOAL, a constraint-aware generative framework that formulates incentive allocation as a conditional sequence generation problem. GOAL directly generates incentive magnitudes conditioned on user histories and system-level global pressure, and integrates a hierarchical causal state encoder to capture both local behavioral dynamics and long-range dependencies. To enable flexible constraint control, we introduce **Safe Constrained Policy Optimization (SCPO)**, which learns a single generative policy that generalizes across a spectrum of ROI constraints without retraining. Experiments on large-scale real-world data and a synthetic fatigue-aware environment show that GOAL improves long-term revenue

and user retention while substantially reducing ROI violation rates compared to strong baselines.

CCS Concepts

• **Information systems** → **Recommender systems**.

Keywords

Autoregressive Generation, Incentivized Advertisement, Policy Optimization

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1 Introduction

Incentivized advertising has become a core mechanism for driving user engagement in modern recommender and advertising systems, where platforms dynamically allocate monetary incentives (e.g., virtual coins or coupons) to stimulate user interactions [5, 44]. Unlike conventional ad recommendation settings that focus solely on content selection [7, 31], incentivized advertising requires jointly deciding how much incentive to allocate for each interaction, under strict global constraints [10, 39] as Fig. 1 shows. Excessive incentives may yield short-term engagement gains but severely harm long-term profitability, while overly conservative strategies fail to

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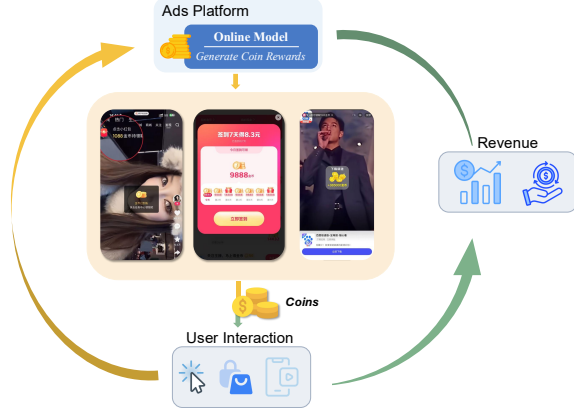


Figure 1: Overview of the incentivized advertising framework. The platform leverages online models to directly generate incentive amounts based on user context and historical behaviors, distributing them to users to encourage participation in ad interactions. Corresponding user interaction histories are collected to enable the platform’s continuous optimization under cost and performance constraints.

sustain user participation. Designing an incentive allocation policy that balances long-term value and financial safety remains a fundamental challenge in large-scale industrial systems.

From a decision-making perspective, incentive allocation can be viewed as a sequential optimization problem with delayed and cumulative effects. Each incentive decision not only affects immediate user response but also influences future behavior through latent dynamics such as user fatigue and incentive saturation. However, existing approaches struggle to address this challenge effectively. Uplift modeling methods, widely adopted in production systems, focus on short-term treatment effects [29, 47, 51] and rely on two-stage pipelines that are highly sensitive to prediction errors, leading to sub-optimal decisions under constraints such as Return on Investment (ROI) [10, 13, 45]. Offline reinforcement learning (RL) methods explicitly optimize long-term objectives, but they typically rely on Markovian assumptions that are violated in high-frequency user interactions [14, 21, 46] and tend to learn overly conservative policies due to distributional shift and limited exploration [18, 19, 25].

Recently, generative sequence modeling paradigms inspired by large language models have shown promise in modeling complex user behaviors by directly generating actions conditioned on historical trajectories [7, 9, 11, 34]. Despite their expressive power, standard Transformer-based architectures exhibit a critical limitation in incentivized advertising [22, 42]: decision-critical signals, such as abrupt user fatigue or intent shifts, are often highly localized in time, while global self-attention tends to dilute these signals [24, 49] when processing long interaction histories. Moreover, existing generative alignment methods primarily optimize unconstrained scalar rewards, making them ill-suited for strict, system-level constraints that couple decisions across time [42].

To address the aforementioned challenges, we introduce **GOAL**, a generative optimization framework with global-level constraints, specifically tailored for incentivized advertising. Our improvements primarily focus on enhancing generative decision models along two

critical dimensions: (i) **capturing non-Markovian and localized user dynamics in long interaction histories**, and (ii) **enforcing strict global-level ROI constraints within a unified generative decision framework**. Concretely, GOAL integrates a hierarchical causal state encoder that explicitly models dense local behavioral dynamics before performing global temporal reasoning, together with a constraint-conditioned generative policy that adapts incentive strategies according to system-level pressure. On top of this architecture, we introduce Safe Constrained Policy Optimization (SCPO), which trains a single generative policy over a distribution of Lagrange multipliers. By separating and normalizing preference signals under each constraint level during alignment, SCPO enables the learned policy to adapt its incentive strategy to different global constraint targets at inference time without retraining. To summarize, our main contributions are as follows:

- **Generative Framework for Incentivized Advertising.** We propose GOAL and explore the application of a generative paradigm to incentivized advertising, enabling the model to capture finer-grained behavioral patterns as well as both short and long-term dependencies.
- **SCPO Alignment Strategy.** We propose SCPO, a constraint-aware alignment algorithm for safe exploration. Unlike GRPO, which may induce *revenue-cost imbalance* by solely maximizing system value, SCPO integrates generative policy optimization with Lagrangian multiplier to strictly enforce global constraints while optimizing long-term ecosystem value.
- **Validation on Industrial Scale.** We validate the superior performance of our approach on a diverse set of real-world industrial and synthetic datasets. The results demonstrate significant improvements in core metrics.

2 Preliminary

In this section, we formally formulate the incentive decision problem as a generative sequence modeling task and introduce the constrained optimization framework based on Lagrangian duality.

2.1 Problem Formulation

Given a dataset $\mathcal{D} = \{(\mathcal{H}_i, \lambda_i, a_i)\}_{i=1}^N$, where N is the dataset size. For the i -th example (corresponding to a specific decision moment t for a user), $\mathcal{H}_i$ represents the historical interaction trajectory derived via a sliding window of length L : $\mathcal{H}_i = [\mathbf{x}_{t-L}, \dots, \mathbf{x}_{t-1}]$, where each composite event $\mathbf{x}_\tau = (\mathbf{s}_\tau, a_\tau) \in \mathcal{S} \times \mathcal{A}$ consists of the high-dimensional state features $\mathbf{s}_\tau$ and the received incentive a_τ at step τ . Additionally, $\lambda_i \in \mathbb{R}^+$ represents the system-level constraint multiplier reflecting the real-time system pressure, and $a_i \in \mathcal{A}$ is the ground-truth incentive allocated at the current step t .

To harness the sequential reasoning capabilities of generative architectures for continuous control, we reformulate the incentive decision as an autoregressive generation task. Specifically, we introduce a discrete vocabulary $\mathcal{V}$, where each token represents a quantized numerical unit. We decompose the continuous target action a_i into a sequence of tokens $\mathbf{g}_i = (g_i^1, g_i^2, \dots, g_i^{T_i})$, where $g_i^k \in \mathcal{V}$ denotes the token at the k -th generation step, and T_i denotes the sequence length. Conversely, we design a deterministic mapping function $\phi(\cdot)$ that reconstructs the original scalar value a_i from $\mathbf{g}_i$, i.e., $a_i = \phi(\mathbf{g}_i) = \sum_{k=1}^{T_i} \phi(g_i^k) \in \mathbb{R}^+$, where $\phi(g_i^k)$ denotes

the numerical magnitude of the token. The details of vocabulary construction and mapping are presented in Section 3.1. Our goal is to train a generative policy π_θ which, given the user history and constraint characteristics $(\mathcal{H}_i, \lambda_i)$, generates the corresponding predicted token sequence $\hat{g}_i = (\hat{g}_i^1, \hat{g}_i^2, \dots, \hat{g}_i^{T_i})$. In turn, the predicted incentive value $\hat{a}_i$, reconstructed via $\hat{a}_i = \phi(\hat{g}_i) = \sum_{k=1}^{T_i} \phi(\hat{g}_i^k)$, approximates the actual incentive a_i while adhering to the cost-efficiency trade-off governed by λ_i .

2.2 Primal-Dual Constrained Policy Optimization

While the generative policy maximizes user value given the system-level state λ , the definition of λ is governed by global business requirements, particularly strict ROI constraints in incentive scenarios.

Primal Problem. Consider a trajectory of N interactions, where action a_i incurs cost c_i and yields revenue r_i . The global ROI is defined as $\frac{\sum_{i=1}^N r_i}{\sum_{i=1}^N c_i}$, which is required to exceed a minimum threshold τ . The resulting reward decision problem is formulated as

$$\max_{\{a_i\}} \sum_{i=1}^N r_i \quad \text{s.t.} \quad \frac{\sum_{i=1}^N r_i}{\sum_{i=1}^N c_i} \geq \tau, \quad 0 \leq c_i \leq C_{\max}. \quad (1)$$

The fractional ROI constraint couples decisions across time, making direct optimization challenging.

Lagrangian Duality and Decoupling. We leverage the Lagrangian duality framework to decouple the global constraint. Specifically, we linearize the ROI constraint and introduce a Lagrange multiplier $\lambda' \geq 0$ to incorporate it into the optimization objective as:

$$\begin{aligned} \mathcal{L}(\pi, \lambda') &= \sum_{i=1}^N r_i + \lambda' \left(\sum_{i=1}^N r_i - \tau \sum_{i=1}^N c_i \right) \\ &= (1 + \lambda') \sum_{i=1}^N r_i - \lambda' \tau \sum_{i=1}^N c_i. \end{aligned} \quad (2)$$

Optimizing this objective is mathematically equivalent to maximizing the normalized return:

$$\max_{\pi \in \Pi} \mathcal{L}(\pi, \lambda) = \max_{\pi \in \Pi} \sum_{i=1}^N (r_i - \lambda c_i), \quad \text{where } \lambda = \frac{\lambda' \tau}{1 + \lambda'}. \quad (3)$$

Under the mild assumption that a feasible policy exists, strong duality holds. Thus, the primal problem is equivalent to the Lagrangian dual: $\min_{\lambda \in \mathbb{R}^+} \max_{\pi \in \Pi} \mathcal{L}(\pi, \lambda)$.

Therefore, given the optimal dual variable λ^* , the optimization objective reduces to:

$$\max_{\pi} \sum_{i=1}^N (r_i - \lambda^* c_i). \quad (4)$$

3 Methodology

As illustrated in Figure 2, we propose GOAL, departing from conventional two-stage incentive optimization pipelines by unifying generation, constraint modeling, and policy optimization into a single generative framework. The model operates in two tightly coupled stages. First, a causal state encoder captures the temporally consistent evolution of user interests, while a Lagrangian multiplier

Algorithm 1: Incentive Vocabulary Construction

Require: Incentive dataset $\mathcal{A} = \{a_j\}_{j=1}^N$, start percentile q_{start} , end percentile q_{end} , decay rate δ , restoration error threshold ϵ_1 , and minimal precision threshold ϵ_2 .

Ensure: Constructed vocabulary $\mathcal{V}$

- 1: Initialize an empty vocabulary $\mathcal{V} \leftarrow \{\}$
- 2: Sort $\mathcal{A}$ in descending order to obtain $\hat{\mathcal{A}} = \{\hat{a}_j\}_{j=1}^N$
- 3: Initialize iteration counter $i \leftarrow 1$, error metric $err \leftarrow \infty$, and current percentile $q \leftarrow q_{\text{start}}$
- 4: **while** $err > \epsilon_1$ **do**
- 5: Compute the q -percentile value o_i of the residual set $\hat{\mathcal{A}}$
- 6: **if** $o_i \leq \epsilon_2$ **then**
- 7: **break** {Terminate if the token precision falls below the minimum threshold}
- 8: **end if**
- 9: Generate a new token w_i such that $\phi(w_i) = o_i$
- 10: $\mathcal{V} \leftarrow \mathcal{V} \cup \{w_i\}$ {Insert the unique token into the vocabulary}
- 11: Update the residual values in $\hat{\mathcal{A}}$:
- 12: **for** $j = 1$ **to** N **do**
- 13: **if** $\hat{a}_j \geq o_i$ **then**
- 14: $\hat{a}_j \leftarrow \hat{a}_j - o_i$
- 15: **end if**
- 16: **end for**
- 17: Update the error metric: $err \leftarrow \max_{1 \leq j \leq N} \left\{ \frac{\hat{a}_j}{a_j} \right\}$
- 18: Update the percentile with decay rate δ :
 $q \leftarrow \max(q \cdot \delta, q_{\text{end}})$
- 19: $i \leftarrow i + 1$
- 20: **end while**
- 21: **return** $\mathcal{V}$

embedding is injected as a global control signal to explicitly parameterize constraint tightness. Second, an autoregressive decoder equipped with constraint-aware MoE routing generates incentive sequences, dynamically switching between conservative and aggressive strategies based on the constraint context. On top of this architecture, we introduce a λ -generalized safe constrained policy optimization scheme (SCPO), enabling a single trained model to adapt to diverse ROI constraints. Together, these components embed constraint awareness into both the representation space and the optimization dynamics, yielding a flexible and scalable solution to constrained incentive generation. In the following subsections, we elaborate on the implementation details of these components.

3.1 Incentive Vocabulary Construction

Incentive allocation differs from language generation in that it involves continuous numerical values with strict magnitude constraints, where existing tokenization [36] or bucketing methods [16, 38] fail to preserve numerical precision. This limitation stems from two inherent characteristics of the action space: a wide value range (e.g., 1 to 3,000 coins) and a heavy-tailed distribution. Standard one-step regression faces a severe optimization bottleneck under such a vast range, as uniform losses struggle to balance low-value regions with high-magnitude scales. Moreover, due to

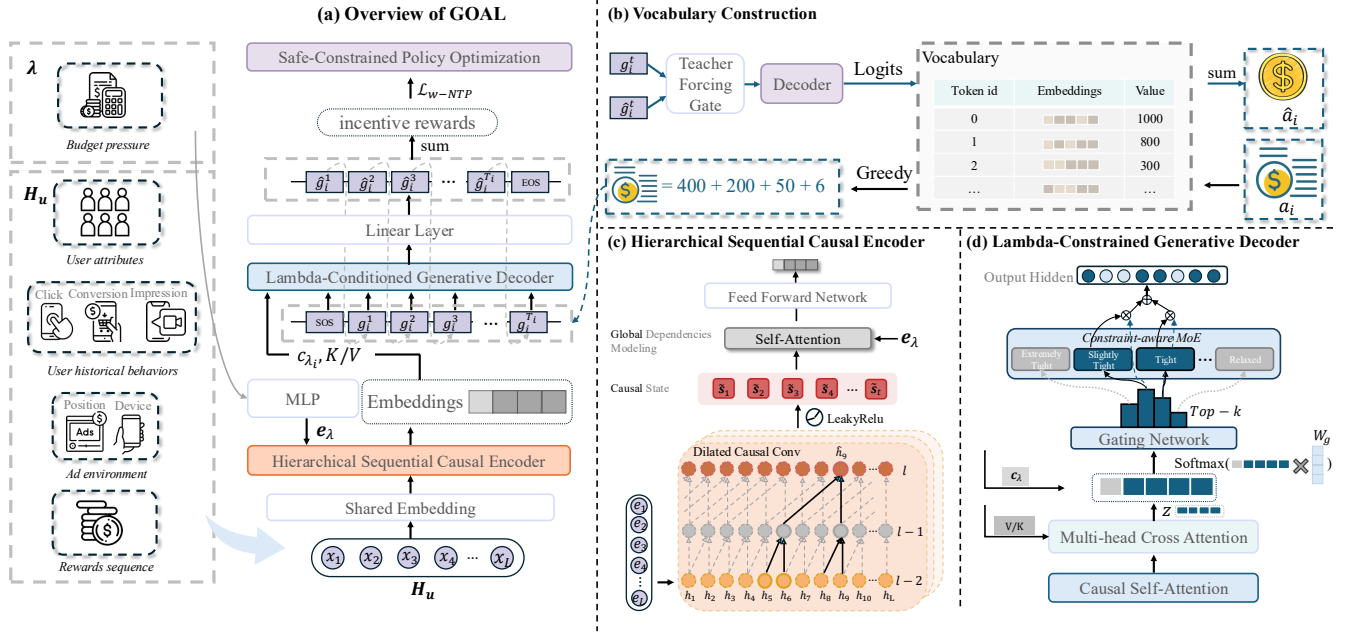


Figure 2: The overall architecture of GOAL. (1) Hierarchical Causal Encoder: synergizes DCC and self-attention to capture both high-frequency local dynamics and global temporal dependencies. (2) λ -Conditioned Generative Decoder: replaces standard FFNs with MoE to enable the experts to perceive different levels of λ -induced constraint pressure. (3) Optimization: optimized via a weighted NTP objective $\mathcal{L}_{W-NTP}$.

the long-tailed nature of marketing traffic, regression models are easily overwhelmed by high-frequency low-value samples, leading to severe underfitting on rare but decision-critical large incentives. To address these challenges, we adopt a data-driven vocabulary construction approach based on dynamic quantile adjustment [27], as detailed in Algorithm 1.

The constructed vocabulary $\mathcal{V} = \{w_1, w_2, \dots, w_{|\mathcal{V}|}\}$ guarantees that each token is unique, enabling the comprehensive representation of the continuous incentive space using a finite set of discrete units. To transform a specific incentive a_i into a token sequence $g_i = (g_i^1, g_i^2, \dots, g_i^{T_i})$, we must ensure that the original value is reconstructible with bounded error while minimizing the sequence length M . To achieve these dual objectives, we employ a deterministic greedy decomposition algorithm that iteratively decomposes the total reward a_i by always selecting the largest available token from $\mathcal{V}$ less than or equal to the current residual value. Crucially, this strategy naturally enforces a monotonicity constraint $\phi(g_i^1) \geq \phi(g_i^2) \geq \dots \geq \phi(g_i^{T_i})$. This captures a hierarchical coarse-to-fine structure, ensuring that the leading tokens encode the dominant magnitude of the incentive, while subsequent tokens efficiently handle residual precision.

3.2 Autoregressive Incentive Generation

The generation module serves as the actor in our framework, which maps the composite state inputs into a precise incentive action.

3.2.1 Hierarchical Causal Encoder. Distinct from conventional encoders that directly feed raw embeddings into Transformer layers, we introduce a specialized causal state encoder module to explicitly

capture the dynamic causality between historical behaviors and current states. The encoding process operates sequentially, moving from initial feature projection to causal modeling, and finally to global context integration.

Given the user historical trajectory $\mathcal{H}_u = [x_{t-L}, \dots, x_{t-1}]$ defined in the problem formulation, which corresponds to the sequence $(x_1, \dots, x_L)$ illustrated in Fig. 2, we first project these features into dense continuous vectors. This process yields an initial embedding sequence $E = (e_1, \dots, e_L)$, which serves as the input to the causal module. To simulate the user's dynamic interests while strictly adhering to temporal causality, we employ a Dilated Causal Convolution (DCC) block [3]. This module aggregates previous behaviors to form a causal state representation. Let $H^{(0)} = E$ be the input to the DCC. We generate the hidden representations $H^{(l)} = (h_1^{(l)}, \dots, h_L^{(l)})$ for the l -th convolutional layer via DCC:

$$H^{(l)} = \text{DilatedCausalConv}(H^{(l-1)}), \quad l = 1, \dots, L_1. \quad (5)$$

Specifically, the hidden state at time t , denoted as $h_t^{(l)}$, is computed by convolving filter weights f with past states:

$$h_t^{(l)} = \sum_{i=0}^{k_c-1} f(i) \cdot h_{t-d_c \cdot i}^{(l-1)}, \quad (6)$$

where d_c and k_c represent the dilation factor and filter size, respectively. This formulation ensures that $h_t^{(l)}$ only depends on historical information, providing strict causality. We further apply residual connections and normalization to enhance gradient flow. Finally, a linear transformation with LeakyReLU activation is applied to

the DCC's output $\mathbf{H}^{(L_1)}$ to obtain the sequence of causal states $\tilde{\mathbf{S}} = (\tilde{s}_1, \dots, \tilde{s}_L)$:

$$\tilde{\mathbf{S}} = \text{LeakyReLU}(\mathbf{W}_s \mathbf{H}^{(L_1)} + \mathbf{B}_s), \quad (7)$$

where $\mathbf{W}_s$ and $\mathbf{B}_s$ are learnable parameters.

To render the generative process ROI-constraint-aware, we introduce a constraint embedding mechanism. Specifically, the scalar Lagrange multiplier λ_i is projected into a dense vector $\mathbf{e}_{\lambda_i}$ via a Multi-Layer Perceptron (MLP), serving as a global condition token: $\mathbf{e}_{\lambda} = \text{MLP}(\lambda_i)$. We prepend this constraint embedding to the causal state sequence, forming an augmented sequence $[\mathbf{e}_{\lambda_i}; \tilde{\mathbf{S}}]$.

While DCC captures local causal dependencies within a finite receptive field, we introduce a self-attention layer to model global dependencies across the entire interaction history, conditioned on λ_i . This enables the capture of long-range behavioral correlations beyond the reach of convolution. The resulting sequence is processed by self-attention followed by an FFN:

$$\mathbf{H}_{enc} = \text{FFN}(\text{SelfAttn}([\mathbf{e}_{\lambda_i}; \tilde{\mathbf{S}}])). \quad (8)$$

The output $\mathbf{H}_{enc}$ provides a semantic representation for the subsequent decoding stage.

3.2.2 λ -Conditioned Generative Decoder. This module formulates incentive generation as an autoregressive sequence generation task. To overcome the precision loss inherent in mapping continuous values to discrete tokens [15, 17, 38], we allow each prediction step the flexibility to choose from the vocabulary.

Let $\mathbf{Z}^{(l)}$ denote the intermediate representation $\mathbf{H}_{enc}$ after attention mechanisms. To enforce the global ROI constraint while maintaining computational efficiency, we substitute the standard Feed-Forward Network (FFN) in the decoder with a constraint-aware Mixture-of-Experts (MoE) layer [8, 50]. For the input vector $\mathbf{z} \in \mathbf{Z}^{(l)}$, the output $\mathbf{o}_{dec}$ is computed by aggregating the outputs of a sparse subset of experts:

$$\begin{aligned} \mathbf{o}_{dec} &= \sum_{j=1}^{N_{exp}} \alpha_j(\mathbf{z}, \mathbf{c}_{\lambda}) \cdot \text{Expert}_j(\mathbf{z}), \\ \mathbf{r} &= \text{Softmax}(\mathbf{W}_g \cdot [\mathbf{z}; \mathbf{c}_{\lambda}]), \\ \alpha_j(\mathbf{z}, \mathbf{c}_{\lambda}) &= \begin{cases} \frac{r_j}{\sum_{e \in \mathcal{T}} r_e}, & j \in \mathcal{T}, \quad \mathcal{T} = \text{Top-}k(\mathbf{r}) \\ 0, & \text{otherwise.} \end{cases} \end{aligned} \quad (9)$$

where N_{exp} denotes the total number of experts, and α_j represents the routing weight for the j -th expert. In this way, the FFN is able to learn different ROI constraints and assign distinct weights to different experts (see Appendix D).

Finally, the MoE output $\mathbf{o}_{dec}$ is projected into the vocabulary space to predict the next token. The generation probability for a token $\hat{g}_i \in \mathcal{V}$ is given by:

$$P_{\theta}(\hat{g}_i^t | \mathbf{H}_{enc}, \hat{g}_i^{<t}) = \text{Softmax}(\mathbf{W}_{out} \cdot \mathbf{o}_{dec}). \quad (10)$$

Weighted Next-Token Prediction Mechanism. Following standard practice, we augment the vocabulary with special tokens $\langle \text{SOS} \rangle$, $\langle \text{EOS} \rangle$, and $\langle \text{PAD} \rangle$ to denote sequence boundaries and enable batch padding. These tokens carry zero incentive value.

As illustrated in Fig. 2, the decoder generates the incentive token sequence $\hat{\mathbf{g}}_i = (\hat{g}_i^1, \dots, \hat{g}_i^{T_i})$ conditioned on the encoder output $\mathbf{H}_{enc}$

and the preceding subsequence $\hat{g}_i^{<t}$. Specifically, at generation step t , the output token $\hat{g}_i^t$ is computed as:

$$\hat{g}_i^t = \arg \max_{g \in \mathcal{V}} P_{\theta}(w | \mathbf{H}_{enc}, \hat{g}_i^{<t}). \quad (11)$$

To equip the model with fundamental market perception and account for the numerical significance of coin values, we employ a Position-Weighted Next Token Prediction objective. In our scenario, tokens in the early stages of the decomposition sequence represent higher orders of magnitude. Therefore, a misprediction at these positions leads to a more substantial bias in the final reward. The objective is defined as:

$$\mathcal{L}_{W-NTP} = -\frac{1}{N} \sum_{i=1}^N \sum_{t=1}^{T_i} \omega_t \cdot \log P_{\theta}(\hat{g}_i^t | \mathbf{H}_{enc}, \hat{g}_i^{<t}) \quad (12)$$

where the weight ω_t is formally defined as follows:

$$\omega_t = \log_{10}(v(g_i^t) + 1) + \delta, \quad (13)$$

where $v(\cdot)$ maps a token to its physical coin value, and δ is a smoothing constant to maintain a baseline gradient for small-value tokens. Distinct from the traditional NTP loss, which treats all token-level prediction errors indiscriminately, our weighted formulation shifts the model's focus toward numerical precision. By penalizing discrepancies based on their physical magnitude, this objective significantly refines the pre-trained model's accuracy in high-value estimation. Meanwhile, to expedite convergence, we also adopt the Teacher Forcing (TF) [35] mechanism, where the ground truth token g_i^t is fed into the decoder at step $t + 1$ to explicitly guide the generation process.

3.3 Reward Model Training

Due to the extreme sparsity of final conversions, using consumption cost alone as the reward signal leads to unstable learning. To provide denser supervision, we incorporate a set of auxiliary behavioral signals $\mathcal{S}_{obj}$ (e.g., revenue contribution, scroll depth, retention, and task-level engagement) to enrich the reward modeling.

To aggregate these heterogeneous objectives, we adopt a neural-based fusion model to learn a personalized Preference Score (P-Score) [4]. The model follows a multi-tower architecture, where each tower encodes objective-specific features, and the resulting representations are fused by a shared MLP to produce a unified utility score. The model is trained with a weighted multi-objective objective to balance trade-offs among different signals:

$$\mathcal{L}_{P\text{-Score}} = \sum_{k \in \mathcal{S}_{obj}} w_k \cdot \mathcal{L}_k(\hat{y}_k, y_k^*), \quad (14)$$

where w_k represents the adaptive weight balancing the trade-off between objectives. For binary classification tasks (e.g., retention or click-through prediction), $\mathcal{L}_k$ is formulated as the standard Binary Cross-Entropy (BCE) loss:

$$\mathcal{L}_k = -\frac{1}{N} \sum_{i=1}^N \left[y_{i,k}^* \log(\hat{p}_{i,k}) + (1 - y_{i,k}^*) \log(1 - \hat{p}_{i,k}) \right], \quad (15)$$

where $y_{i,k}^* \in \{0, 1\}$ denotes the ground-truth label for sample i on objective k , and $\hat{p}_{i,k}$ is the predicted probability.

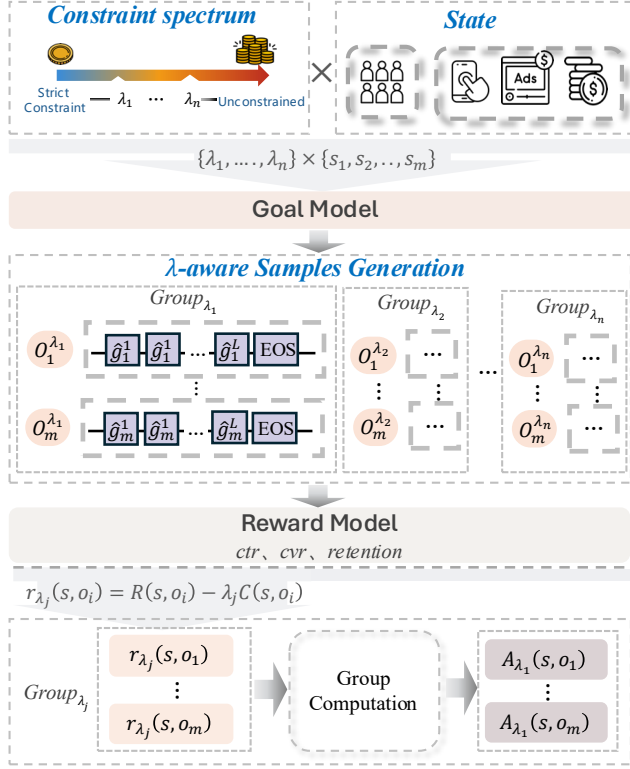


Figure 3: The workflow of Safe Constrained Policy Optimization (SCPO).

3.4 Safe Constrained Policy Optimization (SCPO)

As formulated in Section 2.2, the optimal policy for the condition-constrained alignment problem can be theoretically derived via the dual problem $\min_{\lambda \geq 0} \max_{\pi} \mathcal{L}(\pi, \lambda)$. However, traditional primal-dual methods require iterative updates of the multiplier λ , necessitating policy re-training whenever the ROI threshold shifts. This is computationally prohibitive for large-scale generative models.

To achieve constraint adaptation, we propose a λ -Generalization framework [44] integrated with Group Relative Policy Optimization (GRPO). We fundamentally reformulate the single-point optimization problem into a universal policy learning problem over the distribution of possible Lagrange multipliers.

Universal Lagrangian Objective. Instead of treating λ as a static hyperparameter, we elevate it to a stochastic state variable. Let $\lambda \sim P(\Lambda)$ be a multiplier sampled from a uniform distribution over a predefined range $[\lambda_{\min}, \lambda_{\max}]$. Complementary to this, the range $[\lambda_{\min}, \lambda_{\max}]$ is chosen to cover both unconstrained and strictly constrained regimes. We detail the process of determining the candidate set for λ in Appendix A. We extend the state space to $\tilde{s} = [s; \lambda]$ and parameterize the policy as $\pi_{\theta}(a|s, \lambda)$.

The optimization objective is transformed to maximize the expected Lagrangian-conditioned reward over the entire λ spectrum:

$$\mathcal{J}_{Gen}(\theta) = \mathbb{E}_{\lambda \sim P(\Lambda)} \left[\mathbb{E}_{\tau \sim \pi_{\theta}(\cdot|\lambda)} \left[\sum_{t=0}^T r_{\lambda}(s_t, a_t) \right] \right], \quad (16)$$

where, consistent with the dual formulation in Eq. 3, the composite step-wise reward r_{λ} dynamically subsumes the cost penalty based on the current λ :

$$r_{\lambda}(s, o_i) = R(s, o_i) - \lambda C(s, o_i) \quad (17)$$

, here, R is the P-Score derived in Section 3.3, and C represents the monetary cost of the action. This formulation incentivizes the model to learn a continuous family of strategies: aggressive revenue maximization when $\lambda \rightarrow 0$, and conservative cost control when λ is large.

After training, the learned policy can be instantiated with different constraint levels without retraining. At inference time, a specific λ^* is selected to satisfy the ROI constraint with the detailed selection procedure provided in Appendix B.

SCPO with ROI-Sensitive Advantage. To achieve a policy adaptable to arbitrary constraint levels via preference alignment, we modify the rollout mechanism of the standard GRPO. Specifically, as illustrated in Fig. 3, given state $s \in \{s_1, s_2, \dots, s_m\}$ and a set of parameters $\{\lambda_1, \dots, \lambda_n\}$ sampled from $[0, \lambda_{\max}]$ (A detailed formulation of the candidate selection strategy for λ is provided in Appendix A.), their Cartesian product is constructed as $\{\lambda_1, \dots, \lambda_n\} \times \{s_1, s_2, \dots, s_m\}$. For each pair (λ_i, s_j) , we first generate a set of candidate responses $\{o_i^{\lambda_j}\}_{i=1}^m$. Correspondingly, utilizing Eq. 17, the reward evaluation expands from the original set $\{r_1, \dots, r_m\}$ to a constraint-conditioned matrix $\{r_1^{\lambda_j}, \dots, r_m^{\lambda_j}\}$, i.e.,

$$\mathcal{R}_{\lambda_j} = \{r_{\lambda_j}(s, o_1), \dots, r_{\lambda_j}(s, o_m)\} \quad (18)$$

in each Group λ_j .

This augmented dataset and reward computation integrate the state, policy, and constraint variable λ , facilitating the post-training optimization of the generative policy $\pi(s, \lambda)$ to derive the optimal strategy under any ROI constraint. Finally, during the inference phase, by simply inputting the target optimal λ^* , the model generates the optimal incentive allocation strategy.

Based on the computed constraint-conditioned reward matrix $\{r_{\lambda_j}^{\lambda_j}\}$, the subsequent step involves estimating the advantage to guide the policy gradient update. However, a critical challenge in training with variable constraint inputs is the significant scale discrepancy in rewards across different pressure levels (λ). A high penalty coefficient naturally suppresses the total reward, which could mislead the optimizer into interpreting high- λ trajectories as poor performance solely due to the stricter constraint rather than the policy's quality. Standard global normalization fails to account for this intrinsic shift.

To address this, we define the advantage $A(s, o_i)$ using a Constraint-Conditional Normalization strategy (Appendix C). This approach strictly isolates the evaluation context: for a specific constraint level λ_j , we compute the advantage statistics solely over the m trajectories generated under that same condition. The group statistics are computed as:

$$\mu_{\lambda_j} = \frac{1}{m} \sum_{i=1}^m r_{\lambda_j}(s, o_i), \quad \sigma_{\lambda_j} = \sqrt{\frac{1}{m} \sum_{i=1}^m (r_{\lambda_j}(s, o_i) - \mu_{\lambda_j})^2}. \quad (19)$$

Consequently, the final advantage is formulated as the standardized score within its constraint group:

$$A(s, o_i) = \frac{r_{\lambda_j}(s, o_i) - \mu_{\lambda_j}}{\sigma_{\lambda_j} + \epsilon}, \quad (20)$$

which effectively decouples the policy's performance from the intrinsic difficulty imposed by the constraint level λ .

The final policy optimization objective is computed by maximizing the following surrogate loss, which incorporates a PPO-style clipping mechanism and a KL-divergence regularization term:

$$\mathcal{L}_{\text{SCPO}}(\theta) = \mathbb{E}_s \left[\frac{1}{mn} \sum_{i=1}^m \sum_{j=1}^n \left(\min(\rho_{i,j} A_{i,j}, \text{clip}(\rho_{i,j}, 1 - \epsilon, 1 + \epsilon) A_{i,j}) - \beta \mathbb{D}_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}}) \right) \right]. \quad (21)$$

where $\rho_{i,j} = \frac{\pi_\theta(o_i|s, \lambda_j)}{\pi_{\text{old}}(o_i|s, \lambda_j)}$ denotes the probability ratio between the current and old policies. ϵ is the clipping hyperparameter that restricts the policy update step size, and π_{ref} represents the reference policy used to prevent excessive deviation during training.

4 Experiments

In this section, we conduct comprehensive experiments to demonstrate the effectiveness of GOAL. Specifically, we explore the following research questions:

- **RQ1:** Does GOAL achieve superior long-term revenue and ROI stability compared to state-of-the-art industrial baselines?
- **RQ2:** Can the proposed framework dynamically adapt to varying ROI constraints without re-training?
- **RQ3:** How do the key components (e.g., Causal Encoder, SCPO) contribute to the overall performance?
- **RQ4:** How does GOAL perform in online deployment compared to existing methods?

4.1 Experimental Settings

Datasets. We evaluate our method on two real-world and synthetic datasets, with the experimental setup as described below. Specifically, we use the initial six days of the data as the training set and designate the final day as the test set.

- **IA:** This dataset is collected from the incentive advertising system of a leading short video platform, covering the period from November 5, 2025, to November 12, 2025. It comprises approximately 130,000 daily active users and 1.84 million daily incentive exposure events.
- **Synthetic IA:** Constructed based on the IA dataset to maintain consistent feature alignment, this simulator introduces explicit user fatigue dynamics to benchmark long-term ROI controllability. It contains 10,000 diverse trajectories generated via a mixed behavior policy (details in Appendix E).

Evaluation Metrics. We adopt a comprehensive set of metrics to assess the proposed framework from the perspectives of business growth, cost efficiency, and long-term sustainability:

- **Average Revenue (REV):** The average monetary revenue accrued by the platform from user transactions following incentive exposure.

- **Return on Investment (ROI):** Defined as the ratio of Total Revenue to Total Incentive Cost ($\text{ROI} = \frac{\text{Revenue}}{\text{Cost}}$). This measures the economic efficiency of the allocation strategy.
- **ROI Violate Rate (RVR):** We propose this new metric to measure constraint satisfaction. It is defined as the percentage of sliding windows in which the realized ROI falls below the target threshold. Lower is better.

4.2 Overall Performance (RQ1)

To demonstrate the comprehensive superiority of GOAL, we compare it with $\lambda=0.5$ against three representative state-of-the-art approaches, covering representative paradigms of offline decision making, including generative sequence modeling methods (DT [6] and CDT [26]), unconstrained offline reinforcement learning (IQL [20]) and constrained reinforcement learning approaches (CAL [40] and TREBI [23]). More details about datasets are shown in Appendix F.

The overall performance comparisons of different approaches on the industrial dataset are reported in Table 1. From the experimental results, we can draw the following key observations: **(1) Superior Performance:** GOAL achieves a substantial improvement in total revenue while simultaneously attaining the highest ROI and a lower RVR, indicating that GOAL is able to effectively exploit incentive opportunities without sacrificing efficiency or constraint stability. **(2) Beyond MDP-Based Optimization:** The generative-based methods consistently outperform MDP-based reinforcement learning approaches across multiple evaluation metrics, clearly demonstrating the advantages of generative approaches in balancing constraint satisfaction and revenue optimization. **(3) Explicit constraint-conditioned generation is crucial for incentivized decision-making:** GOAL consistently outperforms CDT and constrained reinforcement learning methods in terms of Revenue, ROI, and RVR, indicating that explicitly conditioning the generative policy on global constraints is a more effective paradigm than post-hoc constrained optimization.

Table 1: Overall performance of GOAL on the IA dataset. The best results are shown in bold, and the second-best results are underlined. Note that the REV is measured in USD.

Method	IA Dataset			Synthetic IA		
	REV ↑	ROI ↑	RVR ↓	REV ↑	ROI ↑	RVR ↓
DT	0.27	<u>4.74</u>	18.23%	0.19	3.64	13.51%
CDT	0.22	4.60	<u>15.00%</u>	0.36	<u>4.12</u>	<u>10.24%</u>
IQL	0.25	3.68	24.69%	0.15	3.59	18.05%
CAL	0.19	3.70	21.82%	0.12	3.61	14.49%
TREBI	0.19	3.92	20.21%	0.09	3.72	14.21%
GOAL (Ours)	0.32	4.93	12.24%	<u>0.29</u>	4.23	9.20%
<i>Improv.</i>	+18.52%	+4.01%	+18.40%	—	+2.40%	+9.86%

4.3 Dynamic Constraint Adaptation (RQ2)

To answer RQ2 regarding dynamic adaptability, we conduct a sensitivity analysis on the Lagrange multiplier λ , which controls the trade-off between incentive cost and revenue. By sweeping λ on the test set, we examine how the learned policy adjusts its behavior under different constraint tightness levels, as shown in Fig. 4.

As illustrated in Fig. 4a, increasing λ leads to a clear and monotonic reduction in the average incentive cost, indicating that higher constraint pressure encourages the policy to adopt more cost-efficient actions. Meanwhile, Fig. 4b shows that the average revenue also decreases as λ increases, reflecting the expected trade-off between revenue maximization and cost control.

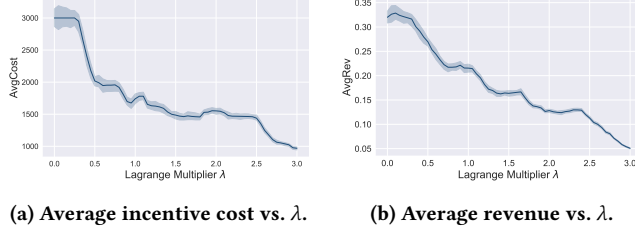


Figure 4: Effect of the Lagrangian multiplier λ on incentive cost and revenue.

In the low- λ regime, the model follows an aggressive strategy by allocating larger incentives to stimulate user engagement, resulting in higher cost and revenue. As λ increases, the policy gradually shifts toward more conservative behavior, selectively reducing incentives for low-efficiency interactions. These results demonstrate that λ serves as an effective inference-time control knob, enabling smooth and predictable adjustment of the policy between aggressive and conservative regimes without retraining.

4.4 Ablation Study (RQ3)

To assess the effectiveness of each component in GOAL, we conducted a series of ablation studies. Specifically, we build several variants of GOAL:

- **w/o SCPO:** A variant of GOAL without the Safe Constrained Policy Optimization stage, which directly uses the supervised fine-tuned (SFT) model to generate incentive tokens.
- **w/o DCC:** A variant of GOAL without the DCC module, which utilizes standard self-attention mechanisms.
- **SCPO w/o λ -generalization:** A variant of GOAL that applies SCPO with a fixed Lagrange multiplier λ , without conditioning the policy on λ .
- **w/o Tokenizer:** A variant that replaces the autoregressive tokenizer with a direct regression head, outputting a single scalar instead of a token sequence under the same configurations.

Table 2: Ablation study on IA dataset. The target ROI threshold is set to $\tau = 3.0$.

Method	REV ($\uparrow$)	ROI ($\uparrow$)	RVR ($\downarrow$)
w/o SCPO	0.27 $\pm$ 0.01	4.57 $\pm$ 0.04	15.64% $\pm$ 0.02
w/o DCC	0.25 $\pm$ 0.01	4.91 $\pm$ 0.05	12.29% $\pm$ 0.02
SCPO w/o λ -gen	0.25 $\pm$ 0.03	4.89 $\pm$ 0.02	14.12% $\pm$ 0.04
w/o Tokenizer	0.29 $\pm$ 0.02	4.89 $\pm$ 0.03	13.85% $\pm$ 0.03
GOAL (Ours)	0.32 $\pm$ 0.01	4.93 $\pm$ 0.02	12.24% $\pm$ 0.01

Overall, the ablation results in Table 2 demonstrate that our model’s effectiveness arises from the complementary roles of its

components. Specifically, the “-w/o SCPO” variant leads to a pronounced degradation in ROI accompanied by an increase in CVR, indicating that the SCPO module is critical for enforcing strict ROI constraints. The “-w/o DCC” variant shows substantial performance degradation across all metrics; relying on standard self-attention, the model is more prone to overfitting spurious correlations in offline data. The “-SCPO w/o λ -gen” variant achieves high ROI with low violation rates but yields the lowest revenue, indicating that a static λ lacks the flexibility to adapt to dynamic system states. Finally, the “w/o Tokenizer” variant shows a clear drop in both REV and ROI. Without structured token generation, the model has difficulty handling the large value range and heavy-tailed distribution of incentive allocations, leading to less accurate modeling of high-stakes cases.

4.5 Online Experimental Testing (RQ4)

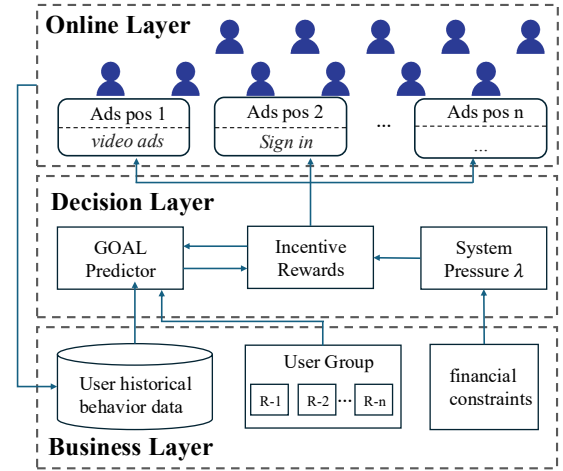


Figure 5: Online Deployment of GOAL

To validate the effectiveness of our model on live traffic, we adapt our framework to the online serving environment, as shown in Fig. 5. We conducted an extended online A/B test on an incentivized advertising system spanning 4 weeks, where each experimental group encompassed 65% of randomly selected users. The control group utilized the production baseline, while the experimental group deployed GOAL. Compared to the baseline, GOAL achieved a **2.184%** increase in ROI and a **2.559%** lift in revenue. Crucially, the ROI improvement remained highly stable over this longer observation window, while secondary metrics such as user dwell time and DAU exhibited no significant adverse changes. These core business gains reached statistical significance with a p -value of 0.03, providing robust evidence of our approach’s long-term effectiveness and financial safety in live commercial systems.

5 Related Works

5.1 Incentive Constraint Rewards Decision

A parallel domain is budget-constrained coupon allocation, which optimizes GMV subject to a fixed budget, distinct from our objective

of maximizing revenue under strict ROI constraints. Mainstream methodologies in this domain fall into two primary categories: Resource Allocation (RA) and Direct Learning Methods (DLM). Approaches based on RA have evolved from disjointed two-stage paradigms [1, 2, 37] to end-to-end Decision-Focused Learning [43, 48]. Works such as [1, 2, 37] typically adopt a predict-then-optimize paradigm, utilizing causal inference models to estimate average treatment effects and subsequently formulating resource allocation as a knapsack problem. In contrast, [43, 48] addresses the non-differentiability of decision losses to achieve the end-to-end integration of prediction and optimization modules. Nevertheless, these approaches are limited by the difficulty of counterfactual inference, compounded by an inherent myopia that neglects future ecosystem value. The second line of research, DLM, bypasses the separation of ML and OR by formulating the problem as a constrained optimization problem, where a substantial body of work [12, 41, 44] frames the coupon allocation task in sequential incentive marketing as a Constrained Markov Decision Process (CMDP) and adopts RL to solve it. These methods typically employ Lagrangian multipliers to incorporate resource constraints directly into the reward function. While capable of modeling long-term rewards, their application to incentivized advertising is hindered by the strict Markovian assumption and the inherent conservatism of offline RL algorithms [18, 19, 25].

Crucially, different from our business scenario, existing methods model incentives as unordered categorical labels. However, in our incentive system (e.g., coin distribution), no such categorical labels are available, making these methods not directly applicable to our precise and fine-grained continuous decision-making task.

5.2 Generative Sequence Modeling and Preference Alignment

The field of user behavior modeling has witnessed a fundamental paradigm shift, transitioning from discriminative matching to generative sequence modeling. Propelled by the triumph of Large Language Models (LLMs), this generative paradigm has gained significant traction. Notably, recent breakthroughs in generative recommendation [7, 9] and search [11, 34] have underscored the immense potential of formulating user modeling as a sequence generation task. To steer these generative paradigms towards specific goals, researchers have adopted Preference Alignment techniques. These methods aim to calibrate the model's generative distribution to match specific reward signals. In pursuit of this alignment, several paradigms have emerged. RLHF [28] typically employs PPO [32] to maximize a learned reward function, yet its reliance on the Actor-Critic architecture often leads to training instability and high computational costs. To mitigate this, DPO [30] derives an analytical solution to implicitly optimize the reward, treating alignment as a stable classification task. Nevertheless, this approach limits the model's ability to explore beyond historical data boundaries. More recently, GRPO [33] estimates advantages via group sampling, eliminating the need for a value network to enhance efficiency. However, these methods predominantly focus on maximizing unconstrained scalar rewards, fundamentally overlooking the non-negligible cost constraints inherent to incentivized advertising.

6 Conclusion

In this paper, we introduced GOAL, the first generative framework designed for incentivized advertising under strict ROI constraints. By unifying causal user state modeling, constraint-aware generation, and SCPO, GOAL addresses the limitations of conventional myopic uplift models and rigid offline RL approaches. Extensive experiments on large-scale industrial datasets demonstrate that GOAL not only outperforms state-of-the-art baselines in maximizing Revenue and user retention but also exhibits superior stability in constraint adherence. Notably, our analysis highlights the efficacy of the proposed Constraint-Aware MoE and λ -generalization mechanism, which empowers the model to adapt to dynamic operational requirements without retraining. This work marks a paradigm shift from traditional prediction-allocation pipelines to a unified generative control process, providing a scalable and flexible solution for complex decision-making in high-stakes commercial environments. In the future, we plan to extend this framework to multi-objective optimization scenarios, such as simultaneously balancing platform revenue, advertiser ROI, and user experience.

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APPENDIX

A Selection of Candidate λ Sets

To enable the policy to perceive and adapt to varying business requirements without retraining, we formalize a Goal-conditioned Policy $\pi(a|s, \lambda)$. In this framework, the state space is augmented with the Lagrangian multiplier λ as a contextual indicator. The objective is to learn a universal strategy manifold that approximates the Pareto frontier between revenue and ROI.

A fundamental challenge is to define an effective sampling interval $[\lambda_{\min}, \lambda_{\max}]$ that covers all potentially optimal strategies. If λ is too small, the policy fails to satisfy ROI constraints; if λ is excessively large, the policy may collapse prematurely. We derive the boundaries based on the marginal utility of actions:

- **Lower Bound** ($\lambda_{\min} = 0$): When $\lambda = 0$, the objective $r_i - \lambda c_i$ reduces to pure revenue maximization. This represents the Unconstrained Boundary, where the policy explores the maximum possible incentive intensity to establish the upper limit of revenue.
- **Upper Bound** ($\lambda_{\max} = \tau$): According to the derived dual relationship $\lambda = \frac{\lambda' \tau}{1 + \lambda'}$ in Eq. 3, the effective penalty coefficient λ asymptotically approaches the ROI threshold τ as the dual penalty $\lambda' \rightarrow \infty$.

To ensure the theoretical rigor of this boundary, we introduce the following property:

Proposition 1. For a discrete action set $\mathcal{A}$ with ascending costs $c_1 < c_2 < \dots < c_k$, there exists a finite threshold $\bar{\lambda} = \max_{i>1} \frac{r_i - r_1}{c_i - c_1}$ such that for all $\lambda \geq \bar{\lambda}$, the optimal action $a = \arg \max_{a_i \in \mathcal{A}} (r_i - \lambda c_i)$ consistently converges to the minimum-cost action a_1 .

Proof. Consider the objective function $\mathcal{F}(\lambda, a_i) = r_i - \lambda c_i$. We aim to find the condition where a_1 is the unique optimizer, i.e., $a_1 = \arg \max_{a_i} \mathcal{F}(\lambda, a_i)$, which must satisfy the system of linear inequalities $r_1 - \lambda c_1 > r_i - \lambda c_i$ for all $i \in \{2, \dots, k\}$. Isolating the terms involving λ yields $\lambda(c_i - c_1) > r_i - r_1$. Given the premise of ascending costs $c_i > c_j$ for $i > j$, the coefficient $(c_i - c_1)$ is strictly positive for all $i > 1$. Dividing both sides by this term defines a set of lower bounds for the multiplier λ as $\lambda > \frac{r_i - r_1}{c_i - c_1}$ for all $i \in \{2, \dots, k\}$. We define the critical threshold $\bar{\lambda}$ as the supremum of these individual action-wise bounds: $\bar{\lambda} = \max_{i \in \{2, \dots, k\}} \left(\frac{r_i - r_1}{c_i - c_1} \right)$. It follows that for any $\lambda \geq \bar{\lambda}$, the inequality $\mathcal{F}(\lambda, a_1) \geq \mathcal{F}(\lambda, a_i)$ is guaranteed for the entire action set $\mathcal{A}$. In this regime, the marginal penalty imposed by λ per unit of cost exceeds the marginal reward gain of any alternative action a_i relative to a_1 . Thus, the optimal policy π^* necessarily converges to the point mass on a_1 . This completes the proof. $\square$

In practical scenarios, business requirements often specify a minimum ROI threshold τ that varies across different marketing campaigns. To ensure our policy can generalize to the most stringent constraints, we identify the maximum value among these potential ROI targets, denoted as $\tau_{\max}$. Since $\tau_{\max}$ typically exceeds the marginal utility threshold $\bar{\lambda}$, we utilize it as the practical sampling ceiling $\lambda_{\max}$. This ensures that the training interval $[0, \lambda_{\max}]$ encompasses all operationally relevant strategy transitions, from unconstrained growth to the most conservative efficiency requirements.

To implement this in our training pipeline, we adopt a discretization and uniform sampling approach. Specifically, the continuous interval $[0, \lambda_{\max}]$ is partitioned into a candidate set $\Lambda = \{0, \Delta\lambda, 2\Delta\lambda, \dots, \lambda_{\max}\}$ with a fixed step size $\Delta\lambda$. For instance, with $\lambda_{\max} = 3.0$ and $\Delta\lambda = 0.5$, the model explores 7 distinct cost-penalty scenarios. During each training iteration, a target λ is randomly drawn from Λ and concatenated with the system state s , enabling the policy $\pi(a|s, \lambda)$ to internalize the mapping between varying ROI constraints and their corresponding optimal decision logics. This stochastic exposure ensures the model approximates a continuous Pareto frontier rather than over-fitting to a single trade-off point.

B Selection of Optimal λ^* at Inference

To ensure the learned policy strictly adheres to the target ROI threshold τ during deployment, we employ a binary search procedure to identify the optimal dual variable λ^* , leveraging the monotonicity of the realized ROI with respect to λ , as detailed in Algorithm 2.

Algorithm 2: λ Calibration

Require: Trained Policy π_θ , Calibration Dataset $\mathcal{D}$, Target ROI threshold τ

Require: Constraint spectrum $\Lambda_{\text{cand}} = \{\lambda_1, \dots, \lambda_n\}$ (sorted in ascending order)

Require: Tolerance ϵ

- 1: Initialize: $low \leftarrow 1$, $high \leftarrow n$, $\lambda^* \leftarrow \lambda_n$ {Default to safety}
- 2: **while** $low \leq high$ **do**
- 3: $mid \leftarrow \lfloor (low + high) / 2 \rfloor$
- 4: $\lambda_{\text{curr}} \leftarrow \Lambda_{\text{cand}}[mid]$
- 5: {Empirical Policy Evaluation}
- 6: Rollout: Generate actions $a \sim \pi_\theta(\cdot | s, \lambda_{\text{curr}})$ for all $s \in \mathcal{D}$
- 7: Calculate Realized ROI: $\hat{\mathcal{R}} \leftarrow \frac{\sum r_i}{\sum c_i}$
- 8: {Binary Search Range Update}
- 9: **if** $\hat{\mathcal{R}} \geq \tau$ **then**
- 10: $\lambda^* \leftarrow \lambda_{\text{curr}}$ {ROI threshold satisfied}
- 11: $high \leftarrow mid - 1$
- 12: **else**
- 13: $low \leftarrow mid + 1$ {ROI threshold violated}
- 14: **end if**
- 15: **end while**
- 16: **return** λ^*

C Safe Constrained Policy Optimization

The algorithmic details of SCPO are presented in Algorithm 3.

D Constraint-aware Experts

To understand how the constraint-aware MoE internalizes the global constraint signal, we analyze the routing behavior of the gating network under different values of the Lagrange multiplier λ . Recall that λ explicitly parameterizes the strength of the ROI constraint and is provided to the gating network as a global conditioning signal.

Fig. 6 visualizes the average expert selection probabilities as λ varies from loose to strict constraint regimes. When λ is small, corresponding to a relaxed ROI constraint, the gating network predominantly activates Expert 0. This expert is consistently assigned the highest routing probability and learns to generate aggressive incentive allocations that prioritize revenue maximization with limited regard for cost efficiency. As λ increases, the routing probability gradually shifts from Expert 0 toward Expert 1 and Expert 2. This smooth transition indicates that the gating network does not treat λ as a discrete switch, but instead continuously modulates the composition of experts to balance revenue and cost under moderate constraint pressure. In the high- λ regime, the routing mass is almost

Algorithm 3: Safe Constrained Policy Optimization (SCPO)**Require:** User state dataset $\mathcal{D}$, Constraint spectrum

$$\Lambda = \{\lambda_1, \dots, \lambda_n\}$$

Require: Policy π_θ , Reference policy π_{ref} , Group size m **Require:** Hyperparameters: clipping ϵ , KL coefficient β , Learning rate η

```

1: while not converged do
2:   Sample a batch of states  $B_s \sim \mathcal{D}$ 
3:   for each state  $s \in B_s$  do
4:     {Candidate Generation}
5:     Sample  $m$  candidate responses  $\{o_i\}_{i=1}^m \sim \pi_\theta(\cdot|s)$ 
6:     {Cartesian Augmentation & Reward Evaluation}
7:     for each constraint level  $\lambda_j \in \Lambda$  do
8:       Compute rewards for all candidates under the
       condition  $\lambda_j$ :
9:        $r_{i,j} \leftarrow R(s, o_i) - \lambda_j C(s, o_i), \quad \forall i \in \{1, \dots, m\}$ 
10:      {Constraint-Conditional Normalization}
11:      Compute group statistics specifically for  $\lambda_j$ :
12:       $\mu_{\lambda_j} \leftarrow \frac{1}{m} \sum_{i=1}^m r_{i,j}, \quad \sigma_{\lambda_j} \leftarrow \sqrt{\frac{1}{m} \sum_{i=1}^m (r_{i,j} - \mu_{\lambda_j})^2}$ 
13:      Compute normalized advantages (Eq. 20):
14:       $A_{i,j} \leftarrow \frac{r_{i,j} - \mu_{\lambda_j}}{\sigma_{\lambda_j} + \epsilon}, \quad \forall i \in \{1, \dots, m\}$ 
15:    end for
16:  end for
17:  {Policy Optimization}
18:  Construct the surrogate objective  $\mathcal{L}_{\text{SCPO}}(\theta)$  over the
  augmented batch:
19:  
$$\mathcal{L} \leftarrow \frac{1}{|B_s| m n} \sum_s \sum_{i=1}^m \sum_{j=1}^n \left[ \min(\rho_{i,j} A_{i,j}, \text{clip}(\rho_{i,j}, 1 - \epsilon, 1 + \epsilon) A_{i,j}) - \beta D_{\text{KL}} \right]$$

20:  Update policy parameters:  $\theta \leftarrow \theta + \eta \nabla_\theta \mathcal{L}$ 
21: end while

```

entirely concentrated on Expert 3, which specializes in conservative, cost-sensitive incentive generation and strongly suppresses low-efficiency actions.

Importantly, this expert specialization is not manually imposed. All experts share the same architecture and are trained jointly, and the observed division of labor emerges naturally from conditioning the MoE on λ during training. These results demonstrate that the constraint-aware MoE effectively translates global constraint pressure into structured, expert-level behavioral specialization, enabling the model to adapt its incentive strategy across different ROI regimes without retraining.

E Synthetic ROI-Constrained Environment

To isolate policy controllability under long-term ROI constraints away from live-data confounders, we introduce a minimal, reproducible synthetic testbed capturing the tension between short-term incentive gains and long-term fatigue-induced ROI degradation. Each session evolves over T timesteps. At step t , the agent observes state $s_t = (f_t, r_{t-1})$, where f_t represents user fatigue and $r_{t-1} \in \{0, 1\}$ is the prior engagement outcome. The agent selects a discrete incentive $a_t \in \{0, 1, \dots, K\}$ with a linear cost $C_t = a_t$.

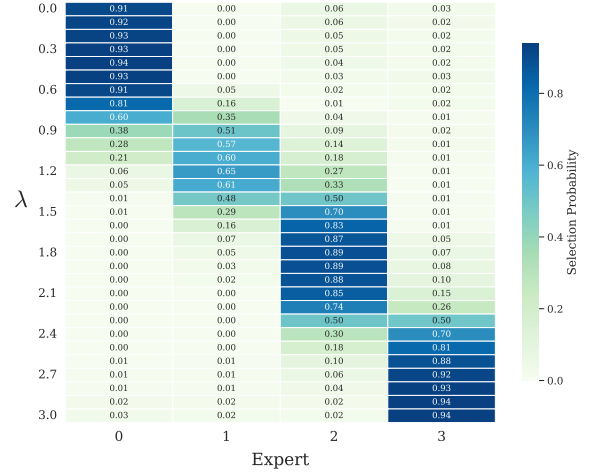


Figure 6: Expert specialization induced by the constraint signal λ in the constraint-aware MoE on IA dataset.

Our fatigue model is intentionally simplified as a controlled testbed for mechanistic verification. Fatigue evolves deterministically as $f_{t+1} = \rho f_t + \eta a_t$, where $\rho \in (0, 1)$ governs natural decay and $\eta > 0$ tracks accumulation. We adopt this exponential decay to align with empirical observations in advertising and recommendation: fatigue from repeated exposures dissipates rapidly at first, then tapers off gradually.

User engagement $r_t \sim \text{Bernoulli}(p_t)$ yields the instantaneous reward $R_t = r_t$. To incorporate saturation effects, fatigue scales user engagement non-linearly via a sigmoid function, yielding the success probability $p_t = \sigma(\alpha a_t - \beta f_t)$, where $\alpha > 0$ dictates the immediate utility of incentives and $\beta > 0$ governs fatigue sensitivity. Trajectory-level ROI is defined as $\text{ROI} = \sum r_t / (\sum a_t + \epsilon)$.

To build the offline dataset, we sample 10,000 trajectories via a mixed behavior policy (greedy, random, and fixed medium incentives) to ensure broad action coverage and fair evaluation. Default parameters are set to $K = 10$, $T = 100$, $\alpha = 0.8$, $\beta = 1.2$, $\rho = 0.9$, and $\eta = 0.5$.

F Dataset Details

- **DT** [6]: A generative approach that casts offline RL as a conditional sequence modeling task, generating actions conditioned on the target return-to-go.
- **CDT** [26]: A safe RL variant of DT that conditions generation on cost thresholds.
- **IQL** [20]: A state-of-the-art offline RL method that mitigates out-of-distribution overestimation via expectile-based value learning.
- **CAL** [40]: It solves the constrained problem by iteratively updating the policy and the dual variable.
- **TREBI** [23]: It tackles constrained policy optimization by optimizing over trajectory distributions while enforcing strict constraint satisfaction.